

Self-Exciting Jumps in the Oil Market: Bayesian Estimation and Dynamic Hedging

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Abstract

In this paper we propose a novel self-exciting jump diffusion model for oil price dynamics based on a Hawkes-type jump-diffusion process. In particular, the jump intensity is stochastic and path dependent, implying that the occurrence of a jump will increase the probability of observing a new jump and this feature of the model aims at explaining the jumps clustering effect. Moreover, volatility is described by a stochastic process, which can jump simultaneously with prices. The model specification is completed by a stochastic convenience yield. In order to estimate the model we apply the two-stage Sequential Monte Carlo (SMC) sampler [22] to both spot and futures quotations. From the estimation results we find evidence of self-excitation in the oil market, which leads to an improved fit and a better out of sample futures forecasting performance with respect to jump-diffusion models with constant intensity. Furthermore, we compute and discuss two optimal hedging strategies based on futures trading. The optimality of the first hedging strategy proposed is based on the variance minimization, while the second strategy takes into account also the third-order moment contribution in considering the investors attitudes. A comparison between the two strategies in terms of hedging effectiveness is provided.

Keywords: Hawkes Processes, Jumps Clustering, Oil Price Dynamics, Particle Filtering, Sequential Monte Carlo, Optimal Hedging.

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1 Introduction

An accurate description of oil price dynamics is crucial for financial applications like risk management, portfolio allocation and derivatives pricing. In addition, the oil market has a strong impact on most aspects of economics in a wider sense; for example, from a macroeconomic perspective oil prices can affect the world GDP growth [27], the efficiency of oil usage and energy consumption [43], and the term structure of interest rates [26]. This explains the huge amount of literature devoted to provide reliable and accurate methods for oil price dynamics calibration and forecast [5].

One of the most popular approaches for modeling commodities is represented by factor models, in which continuous time stochastic differential equations describe the *factors* moving the price dynamics. In particular, among the others we recall the two factor model proposed by Gibson and Schwartz [24], where the spot price is described by a Geometric Brownian Motion (GBM) and the convenience yield by a Ornstein-Uhlenbeck (OU) process. Later, Ribeiro and Hodges [41] proposed a multi-factor model, in which the convenience yield is driven by a Cox-Ingersoll-Ross (CIR) process. This ensures that the convenience yield does not take negative values and hence the model excludes arbitrage. None of these models accounts neither for stochastic volatility nor for jumps.

In order to provide a better description of oil prices dynamics, Larsson and Nossman [32] introduced jumps in both stochastic volatility and returns. They analysed daily spot prices of WTI crude oil from 1989 to 2009, and exploited a Markov Chain Monte Carlo (MCMC) method for the model estimation. In contrast to the literature mentioned before, they did not consider mean reversion in the spot price and also the convenience yield was not introduced as a stochastic factor (they only considered a constant convenience yield in the risk-neutral specification of the model). A similar framework has been adopted also by Brooks and Prockopczuk [7].

Another stochastic volatility model with jumps for oil prices can be found in two recent papers by Fileccia and Sgarra in [18, 19], where the authors included also a stochastic convenience yield. In their paper information from futures prices is included and the model is estimated under both the historical and risk-neutral measure via Particle Markov Chain Monte Carlo (PMCMC).

In our paper we address the problem of describing oil prices dynamics by implementing a different modeling strategy. In particular, we want to investigate if the inclusion of self-exciting effects provides a better understanding of price movements. Self-exciting features have been al-

ready systematically investigated in several asset classes. Fulop et al. [23] estimate a model that considers co-jumps between prices and volatility and self-exciting jump clustering, on the S&P 500 index data from 1980 to 2012; Ait-Sahalia et al. [1] model financial contagion with mutually exciting jump processes and Maneesoonthorn et al. [37] extend the latter model by introducing self-exciting jumps in volatility in a univariate framework, i.e. no mutual excitation between different assets is considered. Finally, we mention the paper by Fulop and Li [22], in which the authors propose a non-affine self-exciting jump diffusion model with stochastic volatility together with a new estimation method, namely the two-stage SMC sampler. Their methodology is applied on the S&P500 and variance swaps observations.

In order to detect self-exciting features in the oil price dynamics we estimate the parameters of a Hawkes-type jump-diffusion model by a particle filtering method. The data set consists of both spot and future quotations of WTI Cushing (Oklahoma) crude oil ranging from January 8, 2008 to December 31, 2018. The particle filtering methodology we apply is based on a hybrid particle filter with a two-stage density tempered Sequential Monte Carlo Method of the same kind of that proposed by Fulop and Li [22]. The model we propose in the present paper exhibits some similarities with the models mentioned above. In particular, our model is similar to the affine version of that considered by Fulop and Li [22], where in addition we introduce another stochastic differential equation, which describes the evolution of the convenience yield as an OU process. However, an interesting feature of the present work is that we apply this kind of jump diffusion model to WTI crude oil spot and futures quotations. Indeed, there is in the literature some intuition behind the existence of this feature also in commodity markets. For example, Filimonov et al. in [20] fit a simple Hawkes process to high frequency data related to many different asset classes, including oil market. Ma et al. [36] consider a plethora of realized range-based volatility models and document an increasing accuracy in futures price volatility forecasting when a Hawkes process is included in the model.

In the present modelling framework jumps with self-exciting features are included in the crude oil spot price dynamics. A remarkable property of the resulting jump-diffusion model is that it is affine, and it allows an explicit computation of the prices of futures contracts as functions of the model parameters, thus providing accurate estimation results. In this paper we address also the issue of hedging against oil price variations. We provide the computation of an optimal dynamic hedging strategy, where optimality refers to an objective function taking into account not only second order effects (variance), but also skewness. The hedging results highlight the

importance of high order effects with respect to the standard approach based only on variance minimization.

The paper is organized as follows. In Section 2 we introduce the Hawkes-type processes and present our self-exciting jump diffusion model. In Section 3 we present our estimation method, while in Section 4 we describe the data set and the results obtained. In section 5 we provide the optimal hedging strategy based on minimization of variance and skewness. In section 6 we provide some concluding remarks.

2 The Model

In this section we introduce the most relevant features of Hawkes processes by following the exposition of Da Fonseca and Zaatour in [12].

The Hawkes process we shall consider throughout the paper exhibits an exponentially decaying intensity, and this is a special kind of point process whose conditional intensity depends on the history of the events. Therefore, let us define in broader terms the conditional intensity function:

Definition 1. Let N_t be a point process and let $\mathcal{F}_t^N$ be the natural filtration generated by N itself. Then, the left continuous process defined by:

$$\lambda(t | \mathcal{F}_{t-}^N) = \lim_{h \rightarrow 0^+} \frac{\mathbb{P}[N_{t+h} - N_t > 0 | \mathcal{F}_{t-}^N]}{h} \quad (1)$$

is called the conditional intensity function of the point process.

Therefore, we have the following

Definition 2. The univariate Hawkes process N with conditional intensity $\lambda(t | \mathcal{F}_{t-}^N) = \lambda_t$, can be defined for all $t > 0$ and $h \rightarrow 0^+$ as:

$$\begin{cases} \mathbb{P}[N_{t+h} - N_t = 1 | \mathcal{F}_{t-}^N] &= \lambda_t h + o(h) \\ \mathbb{P}[N_{t+h} - N_t > 1 | \mathcal{F}_{t-}^N] &= o(h) \\ \mathbb{P}[N_{t+h} - N_t = 0 | \mathcal{F}_{t-}^N] &= 1 - \lambda_t h + o(h) \end{cases} \quad (2)$$

In particular, we can represent the intensity of a Hawkes process by a stochastic differential equation (SDE):

$$d\lambda_t = \beta(\lambda_\infty - \lambda_t)dt + \alpha dN_t, \quad (3)$$

where, $\alpha \in \mathbb{R}^+$ is the magnitude of self-excited jump, $\beta \in \mathbb{R}^+$ is the constant rate of decay and $\lambda_\infty \in \mathbb{R}^+$ is the so called background intensity. Then, by applying Itô formula to $f(t, \lambda_t) = e^{\beta t} \lambda_t$, we obtain the solution to the above SDE:

$$\begin{aligned} \lambda_t &= \lambda_\infty + (\lambda_0 - \lambda_\infty)e^{-\beta t} + \int_0^t \alpha e^{-\beta(t-s)} dN_s, \\ &= \lambda_\infty + (\lambda_0 - \lambda_\infty)e^{-\beta t} + \sum_{j:t_j < t} \alpha e^{-\beta(t-t_j)}. \end{aligned} \quad (4)$$

The solution given by (4) allows to simulate the intensity process. Indeed, different simulation methods are proposed in the literature; for example, one of the most popular procedures is the Ogata's modified thinning algorithm [38], which is based on the thinning method developed by Lewis and Shedler in [34] and further modified by Delay and Vere-Jones [14]. However, more recently Dassios and Zhao in [15] provided an "exact simulation" algorithm, which is much faster than the previous approach. We can see an example of a simulated path of λ_t in Figure 1, where the simulation is performed by applying the algorithm of Dassios and Zhao.

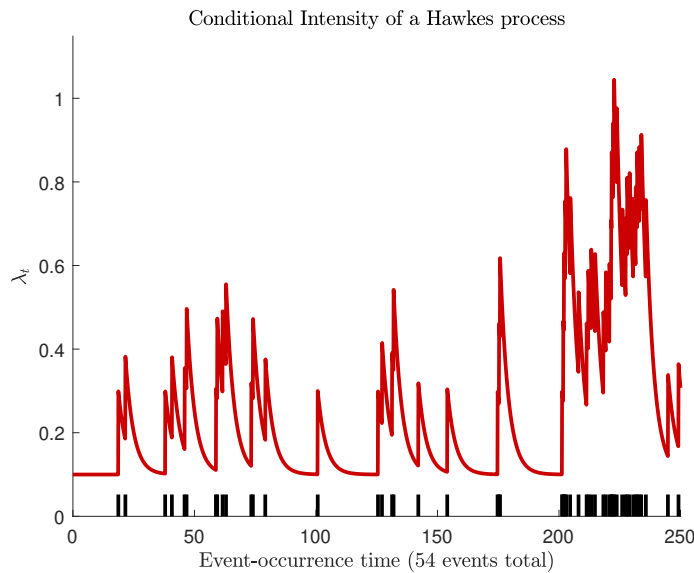


Figure 1: Simulated conditional intensity of a Hawkes process on $[0, T]$ with parameters: $T = 250$, $\lambda_0 = \lambda_\infty = 0.1$, $\alpha = 0.2$ and $\beta = 0.3$.

From figure 1 we can clearly see the clustering of jumps exhibited by the Hawkes process; for

example, in the interval [200, 240] the self-excitation effect looks quite relevant.

An important property of the Hawkes process under consideration is that, although λ_t is clearly non-Markovian, it can be proved that the two-dimensional process (N_t, λ_t) is jointly both a Markov and an affine process and this property improves significantly the analytical tractability of the model. In particular, in [12] the infinitesimal generator and the Dynkin's formula can be used in order to find some moments of the process as solutions of ordinary differential equations.

2.1 Model Dynamics Under The Historical Measure

Now, we can introduce our dynamic model, which takes into account co-jumps between stock price and volatility, stochastic convenience yield and self-excitation. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a complete filtration $(\mathcal{F}_t)_{t \geq 0}$, then our model for $x_t = \ln(S_t/S_0)$ is described by the following system of SDEs:

$$dx_t = \left(\mu - \frac{1}{2}V_t - \lambda_t \mathbb{E}[e^{J_x} - 1] - \delta_t \right) dt + \sqrt{V_t} dW_t + dJ_{x,t}, \quad (5)$$

$$dV_t = k(\bar{V} - V_t)dt + \sigma_v \sqrt{V_t} dW_{v,t} + dJ_{v,t}, \quad (6)$$

$$d\delta_t = \gamma(\bar{\delta} - \delta_t)dt + \sigma_\delta dW_{\delta,t}, \quad (7)$$

$$d\lambda_t = \beta(\lambda_\infty - \lambda_t)dt + \alpha dN_t, \quad (8)$$

From equation (5) we see that changes in the underlying are given by a standard Brownian motion W_t and a compound Poisson process $J_{x,t}$, where the number of jumps N_t is an Hawkes process with stochastic intensity λ_t . Furthermore, the amplitude of jumps is dictated by i.i.d Gaussian random variables with mean μ_J and variance σ_J^2 . On the other hand, Equation (6) describes the evolution of the volatility, which is a mean reverting jump-diffusion process, where $W_{v,t}$ is a standard Brownian motion, possibly correlated with W_t , i.e. $\text{Corr}(dW_1, dW_v) = \rho_v dt$. This feature of the model is important in order to capture the so called *leverage effect*. Furthermore, recent studies find the presence of co-jumps of prices and volatility, not only on the equity market ([16], [17], [22]), but also on the commodity market ([32], [7]). Moreover, there is evidence of jump clustering ([1], [23]), i.e. an extreme movement tends to be followed by another extreme movement. As a consequence, we introduce jumps in volatility with $J_{v,t}$, which is a compound Poisson process with counting process N_t . To be more precise, returns and volatility jump together with the same self-exciting intensity λ_t and the jump size of volatility follows an

exponential distribution with mean μ_v . Finally, since we are dealing with oil prices, there is an additional SDE, which describes the evolution of the convenience yield with a standard Ornstein-Uhlenbeck process as in [24, 42, 31, 44]. The convenience yield includes both the reduction in cost of acquiring inventory and the value of being able to profit from temporary local shortage of the commodity [44], so it is natural to adopt a stochastic process, which could assume both positive and negative values.

In our empirical application we consider two nested models:

- Model I defined by Equations (5)-(8),
- Model II without self-exciting effect, i.e. $\beta = \alpha = 0$.

2.2 Risk-Neutral Dynamics and Futures Pricing

In order to perform joint estimation using spot and futures data, we need to derive a pricing formula for futures contracts. As usual, we employ a suitable change of measure from the real world measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$. To this end consider the following Radon-Nikodym derivative:

$$\frac{d\mathbb{P}}{d\mathbb{Q}} \Big|_{\mathcal{F}_t} = \exp \left\{ - \int_0^t r_s ds - \frac{1}{2} \int_0^t \varphi_x^2(u) du - \int_0^t \varphi_x(u) dW_u - \frac{1}{2} \int_0^t \varphi_\delta^2(u) du - \int_0^t \varphi_\delta(u) dW_u \right\}, \quad (9)$$

where, r_t is the risk-free interest rate. Actually we are pricing only the convenience yield risk, neither volatility nor jump risk. This choice is motivated by the fact the futures price is not a function of spot volatility, jumps or their associated parameters¹ (see [44]). However, volatility and jumps are important in order to describe appropriately spot prices and for hedging purpose, as we shall see later. Thus, following the literature (eg [45], [39], [22]) we left $\varphi_x(t)$ unspecified and choose the convenience yield risk premium as follows:

$$\varphi_\delta(t) = \frac{\varphi_\delta}{\sigma_\delta} \quad \Rightarrow \quad dW_{\delta,t}^{\mathbb{Q}} = dW_{\delta,t}^{\mathbb{P}} + \frac{\varphi_\delta}{\sigma_\delta} dt. \quad (10)$$

¹The economic justification for this result relies in the linearity of the futures payoff. Indeed, volatility and jumps affect higher order moments, but not the first one. From the mathematical point of view, the payoff of the contract is just the expected spot price under $\mathbb{Q}$; thus in the spot price dynamics we compensate the drift with the jump compensator $\lambda_t \mu^*$ and the Itô term $1/2V_t$. As a consequence the solution of the respective ODEs is equal to zero.

Thus, under the risk-neutral measure the structure of the model is preserved:

$$dx_t = \left(r - \frac{1}{2}V_t - \lambda_t \mathbb{E} [e^{J_x} - 1] - \delta_t \right) dt + \sqrt{V_t} dW_t + dJ_{x,t}, \quad (11)$$

$$dV_t = k(\bar{V} - V_t)dt + \sigma_v \sqrt{V_t} dW_{v,t} + dJ_{v,t}, \quad (12)$$

$$d\delta_t = \gamma(\bar{\delta}^{\mathbb{Q}} - \delta_t)dt + \sigma_\delta \sqrt{\delta_t} dW_{\delta,t}^{\mathbb{Q}}, \quad (13)$$

$$d\lambda_t = \beta(\lambda_\infty - \lambda_t)dt + \alpha dN_t, \quad (14)$$

where, $\bar{\delta}^{\mathbb{Q}} = \bar{\delta} - \varphi_\delta/\gamma$.

The model we are proposing belongs to the class of affine models, which means that we know the characteristic function in closed-form. This feature of affine models is very important in order to price financial derivatives as we shall see below. The payoff of a futures contract $F(t, \tau)$, with time to maturity $\tau = T - t$, is given by the usual relation:

$$F(t, \tau) = \mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[e^{x_T} \mid \mathcal{F}_t], \quad (15)$$

where $x_T = \ln(S_T)$. We start by considering the joint moment generating function (MGF) of all state variables:

$$G(w, x_t, V_t, \delta_t, \lambda_t, t, \tau) = \mathbb{E} [e^{wx_T}]$$

Now, since under the risk-neutral measure the model structure is the same we drop the $\mathbb{Q}$ superscript to lighten the notation, and by Feynman-Kac theorem we obtain the following PDE:

$$\begin{aligned} & -G_\tau + \left(r - \frac{1}{2}V_t - \lambda_t \mathbb{E} [e^{J_x} - 1] - \delta_t \right) G_x + \frac{1}{2}V_t G_{xx} + k(\bar{V} - V_t)G_v + \\ & + \frac{1}{2}\sigma_v^2 V_t G_{vv} + \rho_v \sigma_v V_t G_{xv} + \beta(\lambda_\infty - \lambda_t)G_\lambda + \gamma(\bar{\delta} - \delta_t)G_\delta + \frac{1}{2}\sigma_\delta^2 G_{\delta\delta} \\ & + \lambda_t \int [G(w, x_t + J_x, V_t + J_v, \delta_t, \lambda_t + \alpha, t, T) - G(w, x_t, V_t, \delta_t, \lambda_t, t, T)] \nu(dJ_x, dJ_v) = 0 \end{aligned} \quad (16)$$

with terminal condition $G_T = \exp(wX_T)$. Now, we guess a solution of the form:

$$G(w, x_t, V_t, \delta_t, \lambda_t, t, \tau) = \exp \{ wx_t + A(w, \tau) + B(w, \tau)V_t + C(w, \tau)\delta_t + D(w, \tau)\lambda_t \} \quad (17)$$

subject to $A(0) = 0$, $B(0) = 0$, $C(0) = 0$ and $D(0) = 0$. From the PDE (16) we obtain the

following system of ODEs:

$$\begin{cases} \frac{\partial A(w, \tau)}{\partial \tau} = rw + \beta \lambda_\infty D(w, \tau) + k \bar{V} B(w, \tau) + \gamma \bar{\delta} C(w, \tau) + \frac{1}{2} \sigma^2 C^2(\tau), \\ \frac{\partial B(w, \tau)}{\partial \tau} = -\frac{1}{2}(w - w^2) - (k - \rho \sigma_v w) B(w, \tau) + \frac{1}{2} \sigma_v^2 B^2(w, \tau), \\ \frac{\partial C(w, \tau)}{\partial \tau} = -w - \gamma C(w, \tau), \\ \frac{\partial D(w, \tau)}{\partial t} = -\beta D(w, \tau) + \int [e^{w J_x + B(w, \tau) J_v + D(w, \tau) \alpha} - 1] \nu(dJ_x, dJ_v) - \mathbb{E}[e^{J_x} - 1] w. \end{cases} \quad (18)$$

Now, the futures price is given simply by the MGF computed in $w = 1$; then the solution of the second ODE is $B(\tau) = 0$, since it is a Riccati equation without the constant term. Due to this result we also have $D(\tau) = 0$. Hence, as we said before, by construction future prices are not affected by volatility and jumps. Therefore, the solution for the log-futures is of the form:

$$\ln F(t, \tau) = \ln S_t + A^{\mathbb{Q}}(\tau) + C^{\mathbb{Q}}(\tau) \delta_t, \quad (19)$$

where,

$$A^{\mathbb{Q}}(\tau) = -\frac{e^{-\gamma\tau} (\delta^{\mathbb{Q}}(-e^{\gamma\tau}) + \delta^{\mathbb{Q}}\gamma\tau e^{\gamma\tau} + \delta^{\mathbb{Q}} - \gamma r \tau e^{\gamma\tau})}{\gamma}, \quad (20)$$

$$C^{\mathbb{Q}}(\tau) = \frac{e^{-\gamma\tau} - 1}{\gamma}. \quad (21)$$

3 Parameters Estimation Method

In order to estimate the parameters using real data, we cast our model in a state-space form and apply a simple Euler scheme. Denoting by Δt a small time interval, the observation equation for stock prices is given by

$$\ln S_t = \ln S_{t-1} + \left(\mu - \frac{1}{2} V_t - \lambda_{t-1} \mathbb{E}[e^{J_x} - 1] - \delta_{t-1} \right) \Delta t + \sqrt{V_{t-1} \Delta t} W_t + J_x \Delta N_t, \quad (22)$$

where, $J_x \sim \mathcal{N}(\mu_J, \sigma_J^2)$, $W_t \sim \mathcal{N}(0, 1)$ and $\Delta N_t = N_t - N_{t-1} \sim \text{Bernoulli}(\lambda_{t-1} \Delta t)$.

In this work we also consider futures prices with n maturities. As usual in the literature, we observe derivatives data with measurement errors ([22], [16], etc.); then we have an additional

observation equation:

$$\ln F(t, T)^O = \ln F(t, T)^M + \epsilon_t,$$

where, $\ln F(t, T)^O$ is a vector of observed futures prices at time t with maturity T , and $\ln F(t, T)^M$ are corresponding prices obtained with Equation (19). The measurement errors follow a multivariate normal, i.e. $\epsilon_t \sim \mathcal{N}(0, \Omega)$, with $\Omega = \Sigma \Sigma^T$ and $\Sigma = \sigma_e I_n$.

Our model features latent states, which cannot be observed. The discretized version of Equations (6)-(7)-(8) reads as follows:

$$V_t = V_{t-1} + k(\bar{V} - V_{t-1})\Delta t + \sigma_v \sqrt{\Delta t} V_{t-1} Z_t + J_v \Delta N_t \quad (23)$$

$$\delta_t = \delta_{t-1} + \gamma(\bar{\delta} - \delta_{t-1})\Delta t + \sigma_\delta \sqrt{\Delta t} W_{\delta,t} \quad (24)$$

$$\lambda_t = \lambda_{t-1} + \beta(\lambda_\infty - \lambda_{t-1})\Delta t + \alpha \Delta N_t \quad (25)$$

where, Z_t is a standard normal correlated with W_t in Equation (22), $J_v \sim \text{Exp}(\mu_v)$ and $W_{\delta,t}$ is an independent standard normal. To address the estimation of the model we rely on Sequential Monte Carlo Methods (SMC). In particular, we shall consider the two-stage density tempered SMC, recently proposed in [22]. This method provides a direct route from the prior to the posterior using all data in a row. To be more precise, by starting from the prior, we temper the distribution in order to gradually reach the final posterior through the tempering coefficient ξ , which is chosen adaptively to ensure satisfactory particle diversity, as in [22]. In this framework we proceed in two steps: in the first stage we use a small number of state particles M_1 in order to provide a coarse exploration of the posterior and then in the second stage we increase the number of particles M_2 to correct the error in the likelihood estimation. Usually, the second stage is much faster than the first one leading to an overall advantage with respect to an SMC sampler which runs throughout with a fixed M_2 . The other main feature of this procedure is the smoothing of the likelihood in the particle filter and the introduction of common random numbers in order to reduce the variance in the likelihood estimation, which is crucial to obtain reliable estimates. Furthermore, our estimation approach delivers an approximation of the marginal likelihood,

$$p(y_{1:T}) = \int p(y_{1:T} | \Theta) p(\Theta) d\Theta,$$

which can be used to construct Bayes factors for model comparison. More precisely, for two models M_1 and M_2 the Bayes factor is given by the ratio of their marginal likelihoods,

$$BF_{1,2} = \frac{p(y_{1:T} \mid M_1)}{p(y_{1:T} \mid M_2)}.$$

The Bayes factor does not rely on asymptotic distribution theory and provides a simple way to evaluating different models. Besides, it contains a penalty for using more parameters. For a complete exposition about the two-stage SMC sampler algorithm we refer to [22].

3.1 An Hybrid Particle Filter

Our estimation strategy is based on Bayesian inference. Hence, if we denote the set of model parameters as Θ and all observations and latent states as $y_{1:T} = \{\ln S_t, \ln F(t, T)^O\}_{t=1}^T$ and $x_{1:T} = \{V_t, \delta_t, \lambda_t\}_{t=1}^T$ respectively, we can define the joint posterior distributions as

$$p(\Theta, x_{1:T} \mid y_{1:T}) = p(x_{1:T} \mid \Theta, y_{1:T})p(\Theta \mid y_{1:T}) \quad (26)$$

The state-space model we are concerned with is clearly non-linear and non-Gaussian, thus standard Kalman filtering techniques cannot be applied. Therefore, we rely on the application of particle filtering methods, which are simulation based methods able to take into account the complexity of our model. The general idea is to approximate continuous time distributions with discrete points, called particles.

Given a set of particles $\{x_{t-1}^{(i)}\}_{i=1}^M$ representing the filtering distribution $p(X_{t-1} \mid y_{1:t-1})$ at time $t-1$, we can decompose the filtering distribution at time t as follows:

$$p(x_t \mid y_{1:t}) \propto \int p(y_t \mid x_t)p(x_t \mid x_{t-1})p(x_{t-1} \mid y_{1:t-1})dx_{t-1}. \quad (27)$$

Now, by importance sampling we can sample from a proposal density $q(x_t \mid x_{t-1})$ and then attach importance weights w_t to account for the difference between the target and the proposal:

$$w_t^{(i)} = \frac{p(y_t \mid x_t^{(i)})p(x_t^{(i)} \mid x_{t-1}^{(i)})}{q(x_t^{(i)} \mid x_{t-1}^{(i)})}, \quad (28)$$

and denote with $W_t^{(i)} = w_t^{(i)} / \sum_{j=1}^M w_t^{(j)}$ the normalized weights. This algorithm is known as sequential importance sampling and suffers from the so called *particle degeneracy*. To solve this

problem we resample the particles proportional to $W_t^{(i)}$, obtaining an equally weighted sample that can be used to approximate the filtering distribution:

$$\hat{p}(x_t \mid y_{1:t}) = \frac{1}{M} \sum_{i=1}^M \delta_{x_t^{(i)}}(\tilde{x}_t^{(i)}), \quad (29)$$

where, $\tilde{x}_t^{(i)}$ are the resampled particles. This is known in the literature as sequential importance resampling (SIR). As a byproduct, the particle filter algorithm delivers an unbiased estimate of the marginal likelihood:

$$\hat{p}(y_{1:t} \mid \Theta) = \prod_{l=2}^T \hat{p}(y_l \mid y_{1:l-1}, \Theta) \hat{p}(y_1 \mid \Theta), \quad (30)$$

where,

$$\hat{p}(y_l \mid y_{1:l-1}, \Theta) = \frac{1}{M} \sum_{i=1}^M w_l^{(i)}. \quad (31)$$

The most common particle filter is the bootstrap filter presented in [25], where the proposal density is simply the state transition law, i.e. $q(x_t \mid x_{t-1}) = p(x_t \mid x_{t-1})$, which does not take into account the new observation y_t , leading to poor performances if the observation is highly informative (e.g. in the case of a jump). In our setting we are not able to derive the optimal proposal, but we can design an hybrid smooth SIR, which is a relatively efficient particle filter with respect to outliers. In particular, to robustify the filter against outliers, we propose jump times ΔN_t from a Bernoulli with probability 0.5. Furthermore (as in [22]) instead of resampling, we fit a multivariate normal distribution on the smoothing distribution and sample from this normal using the inverse CDF method. The detailed algorithm is outlined below.

At time t we have from time $t-1$ a weighted sample of M particles representing the filtering distribution: $x_{t-1}^{(i)} = \{V_{t-1}^{(i)}, \lambda_{t-1}^{(i)}, \delta_{t-1}^{(i)}\}_{i=1}^M$.

Hence, for each particle, the PF works as follows:

- **Step 1 (Sample $\Delta N_t^{(i)}$ from proposal):** $\Delta N_t^{(i)} \sim \text{Ber}(0.5)$. Then, sample return jumps $J_x^{(i)} \sim \mathcal{N}(\mu_J, \sigma_J^2)$, variance jumps $J_v^{(i)} \sim \text{Exp}(\mu_v)$ and the latent states from their

transitions:

$$\begin{aligned}\lambda_t^{(i)} &= \lambda_{t-1}^{(i)} + \beta(\lambda_\infty - \lambda_{t-1}^{(i)})\Delta t + \alpha\Delta N_t^{(i)} \\ V_t^{(i)} &= V_{t-1}^{(i)} + k(\bar{V} - V_{t-1}^{(i)})\Delta t + \sigma_v\sqrt{\Delta t V_{t-1}^{(i)}} \left(\rho_v Z_t^{(i)} + \sqrt{1 - \rho_v^2} W_{v,t}^{(i)} \right) + J_v^{(i)} \Delta N_t^{(i)} \\ \delta_t^{(i)} &= \delta_{t-1}^{(i)} + \gamma(\bar{\delta} - \delta_{t-1}^{(i)})\Delta t + \sigma_\delta\sqrt{\Delta t} W_{\delta,t}^{(i)}\end{aligned}$$

where, $W_{v,t}^{(i)}$ and $W_{\delta,t}^{(i)}$ are independent normals and

$$Z_t^{(i)} = \frac{\ln S_t - \ln S_{t-1} - \mu_x \Delta t - J_x^{(i)} \Delta N_t^{(i)}}{V_{t-1}^{(i)} \Delta t},$$

with, $\mu_x = \mu - \frac{1}{2}V_t - \lambda_{t-1}\mathbb{E}[e^{J_x} - 1] - \delta_{t-1}$.

- **Step 2 (Reweight):** Compute log-weights according to

$$\log p(\ln S_t \mid \ln S_{t-1}, x_{t-1}, \Delta N_t) + \log p(\ln F_{t,T} \mid \ln S_t, \delta_{t-1}^{(i)}) + \log \pi_t \rightarrow \log w_t^{(i)}$$

where,

$$\begin{aligned}\text{if } \Delta N_t^{(i)} = 0 &\Rightarrow \log \pi_t = -\lambda_t^{(i)} \Delta t - \log(0.5) \\ \text{if } \Delta N_t^{(i)} = 1 &\Rightarrow \log \pi_t = -\log(1 - \exp(\lambda_t^{(i)} \Delta t)) - \log(0.5)\end{aligned}$$

- **Step 3 (Smooth approximation):** Generate from $x_{t-1|t} = \{V_{t-1|t}, \delta_{t-1|t}, \lambda_{t-1|t}\}$ fitting a multivariate normal. As suggested in [22] we add a moment-matching step.

4 Empirical Application

4.1 Data Description

In this section we present the data we are going to examine in our empirical analysis. In particular, we consider WTI Cushing (Oklahoma) crude oil spot and futures quotations obtained from Bloomberg². In Figure 2 we plot a time series of daily log-returns: $Y_t = \log(S_t/S_{t-1})$, ranging from 08/01/2008 to 31/12/2018, thus 2767 spot data are considered. The data cover some crucial historical periods as the global financial crisis, when oil prices experienced huge

²spot data ticker: USCRWTIC, futures data ticker: CLm, for $m = \{3, 6, 9, 12, 18, 21, 24\}$.

fluctuations between 2008 and 2010. Moreover, in the middle of 2014, price started declining due to a significant increase in oil production in USA, and declining demand in the emerging countries. Besides, from 2016 to 2018, complex negotiations with OPEC led to higher variability in oil prices.

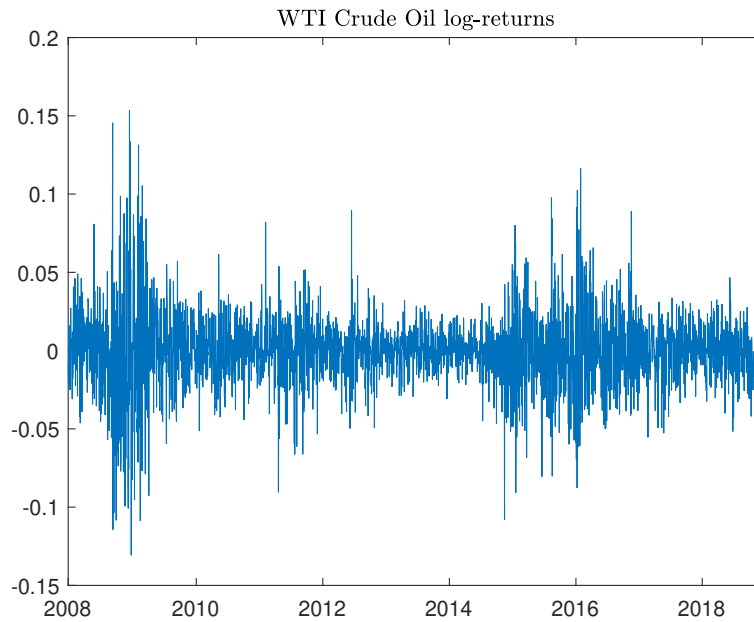


Figure 2: WTI crude oil log-returns from 08/01/2008 to 31/12/2018.

First, a preliminary statistical analysis is performed on log-returns. To this end in Table 1 we show some descriptive statistics and the result of a Jarque-Bera (JB) normality test. In addition a QQ-plot is provided in Figure 3. This simple analysis suggests that observations do not come from a Gaussian distribution.

Statistics	Log-return
Observations	2767
Mean	-2.7179e-04
Standard Deviation	0.0245
Skewness	0.1181
Kurtosis	7.3057
Min value	-0.1307
Max value	0.1533
JB test	Rejected

Table 1: Descriptive statistics and JB test on WTI crude oil log returns observed with daily frequency over the period 08/01/2008 to 31/12/2018.

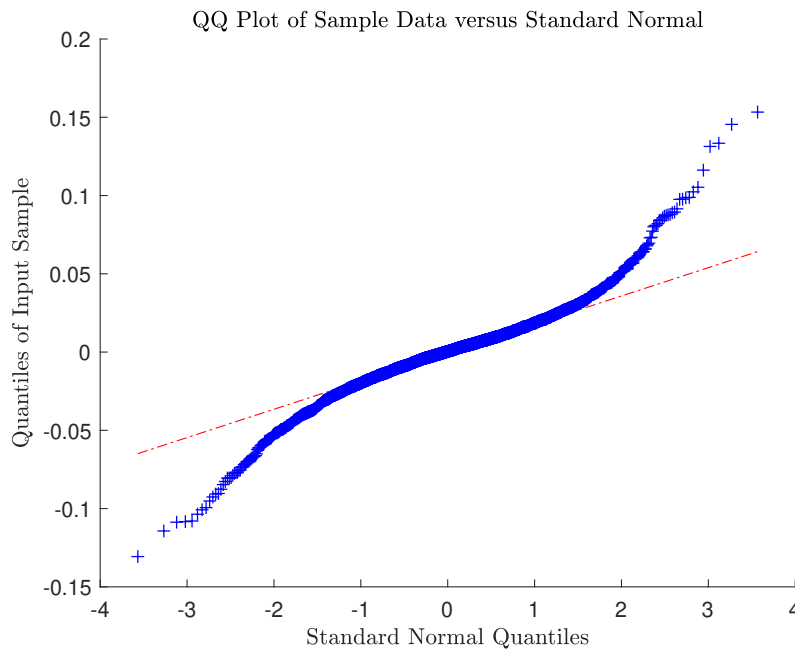


Figure 3: Q-Q Plot for daily frequency returns on WTI crude oil.

On the other hand, we consider futures contracts with respect to WTI crude oil. In Figure 4 we show log-futures prices from 08/01/2008 to 31/12/2018, for maturities τ equal to 3, 12 and 24 months. For estimation purpose, we retain 8 futures contracts with fixed maturities

ranging from 3 to 24 months. Contracts with more than 2 years of maturity are less liquid [31], therefore they are not considered in our estimation procedure. As for spot observations, we conduct a statistical analysis on the whole futures data in Table 2.

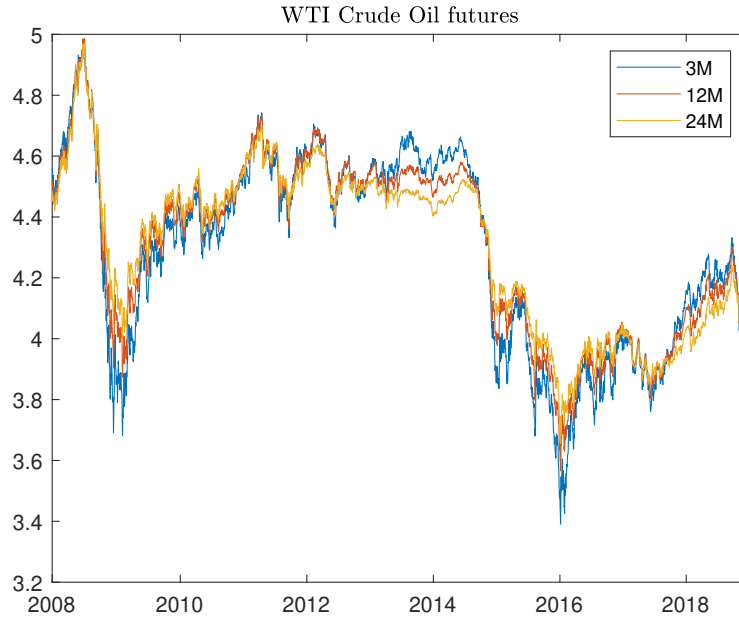


Figure 4: WTI futures contracts from 08/01/2008 to 31/12/2018.

Statistics	CL3	CL6	CL9	CL12	CL15	CL18	CL21	CL24
Mean	4.2842	4.2968	4.3019	4.3041	4.3047	4.3046	4.3043	4.3039
Standard Dev	0.3217	0.3058	0.2956	0.2879	0.2814	0.2760	0.2713	0.2673
Skewness	-0.3142	-0.2751	-0.2547	-0.2422	-0.2337	-0.2277	-0.2224	-0.2201
Kurtosis	2.0059	1.9591	1.9465	1.9358	1.9328	1.9388	1.9475	1.9598
Min value	3.3908	3.4825	3.5293	3.5656	3.5943	3.6168	3.6368	3.6548
Max value	4.9845	4.9894	4.9895	4.9858	4.9822	4.9782	4.9747	4.9716

Table 2: Descriptive statistics on WTI crude oil log futures prices observed with daily frequency over the period 08/01/2008 to 31/12/2018.

4.2 Estimation Results

In this subsection we present the results obtained from the estimation procedure. In our empirical application, for the two-stage SMC sampler, we set the number of parameter particles N equal to 1000, the number of state particles at first stage M_1 equal to 30, and the number of state particles M_2 at second stage equal to 500. The algorithm is initialized using the priors in Table 3 and with $\Sigma = \sigma_e I_n$. The choice of the hyper-parameters of the prior distributions is based on calibration using the training sample approach, which is widely used to calibrate the objective priors ([21, 22, 23]). Notably, we find that most parameters are not sensitive to the selection of the priors. Moreover, we take the random walk proposal, trigger the resample-move step when the ESS reaches $N/2$, and then keep moving until the cumulative average acceptance rate across the population reaches 2. Finally, we find hard to estimate all the parameters governing the jump intensity. This phenomenon is not surprising using only underlying data, as observed by [23]. Therefore, in order to alleviate the identification problem, we fix one those parameters³. In particular, a natural solution is to set λ_∞ equal to the value usually obtained in the literature in model with constant jump intensity, which is approximately equal to 6 ([32, 18, 19, 7]).

Θ	Dist	Support	(μ_0, σ_0)	Θ	Dist	Support	(μ_0, σ_0)
μ	Normal	$(-\infty, \infty)$	(0.02, 0.15)	μ_v	Tr. Normal	$(0, \infty)$	(0.02, 0.1)
μ_J	Tr. Normal	$(0, \infty)$	(-0.02, 0.08)	γ	Normal	$(0, \infty)$	(0.9, 0.5)
σ_J	Tr. Normal	$(0, \infty)$	(0.05, 0.1)	$\bar{\delta}$	Tr. Normal	$(0, \infty)$	(0.15, 0.05)
k	Tr. Normal	$(0, \infty)$	(3.0, 4.0)	σ_δ	Tr. Normal	$(0, \infty)$	(0.25, 0.1)
$\bar{V}$	Tr. Normal	$(0, \infty)$	(0.05, 0.06)	α	Tr. Normal	$(0, \infty)$	(7.0, 8.0)
σ_v	Tr. Normal	$(0, \infty)$	(0.25, 0.25)	β	Tr. Normal	$(0, \infty)$	(10.0, 8.0)
ρ_v	Tr. Normal	$[-1, 1]$	(-0.7, 0.5)	φ_δ	Tr. Normal	$(-\infty, \infty)$	(0.5, 0.1)

Table 3: Priors. The prior related to the pricing errors is given by a multivariate normal, where the elements of the mean vector are all equal to 0.01, and the diagonal covariance matrix is $\sigma_e I_8$, with $\sigma_e = 0.05$

To appreciate the efficiency of our estimation method, we can have a look at the acceptance rates related to the moving step in Figure 5. The star-line indicates the acceptance rates from the first stage and the circle line those from the second stage. We clearly see that the acceptance

³We postpone to another paper the inclusion of highly informative observations like options.

rates remain high during both the first and second stage. Furthermore, as demonstrated in [22], the number of density-bridging iterations is much smaller in the second stage with respect to the first stage.

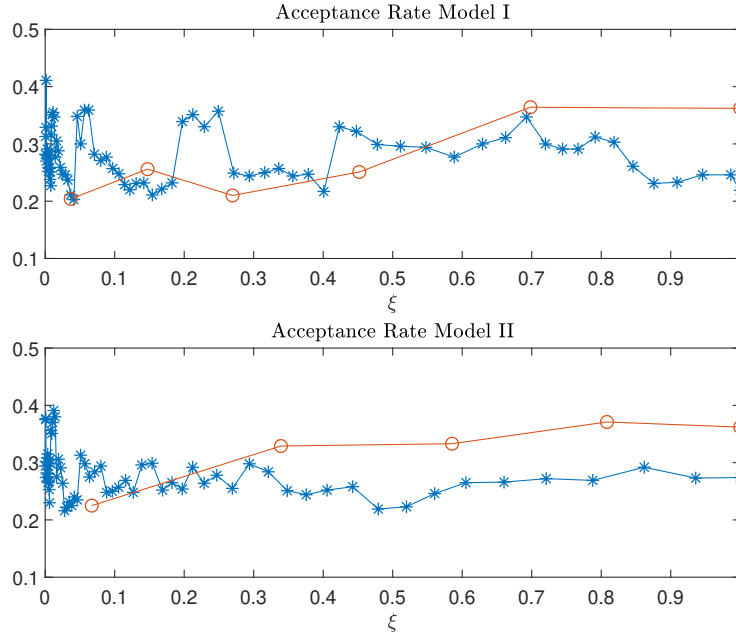


Figure 5: The figure plots the last acceptance rate in moving steps at each density-bridging iteration with respect to ξ_i for model I (upper) and model II (bottom). In the algorithm, ξ is automatically selected using a grid search approach. The blue line refers to the first stage, while the orange line refers to the second stage.

We now move to the posterior distributions of the fixed parameters of Model I, which are shown in Figure 6. From the plots we assess the degree of learning of our method and we notice that the only prior which is slightly informative is the vol-of-vol σ_v .

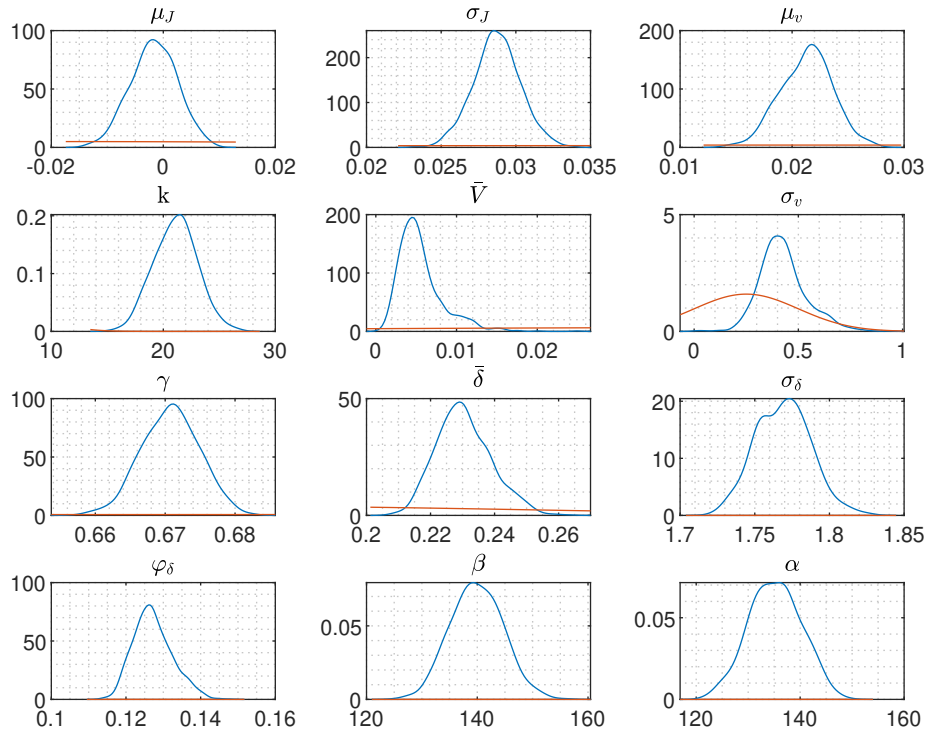


Figure 6: The figure presents for model I the posterior kernel density estimation of the parameters (blue line) compared to their priors (red line).

Figures 7 and 8 plot the filtered volatility, convenience yield and jump intensity obtained by running our smooth particle filter at the posterior mean. In particular, volatility is quite persistent and in periods when prices fall down we observe a rise in volatility according to the well known leverage effect. The convenience yield is moving in the same direction of oil prices, which is consistent with its definition and provides a clear economic intuition. A large amount of literature devoted to commodities provides a confirmation and an explanation of this behavior, we just mention [3] and [33] among many other contributions on this topics, some based on Normal Backwardation Theory [35], some on Theory of Storage [9]. The intensity process present a self-exciting behaviour; this will be confirmed later when focusing on the parameters estimates.

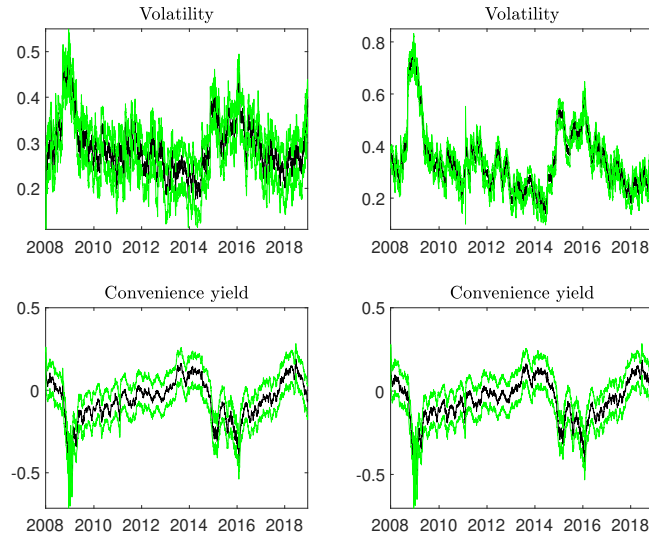


Figure 7: The figure presents the filtered volatility and convenience yield from model I (left) and II (right). The posterior mean and (5, 95)% quantiles are reported at each time point.

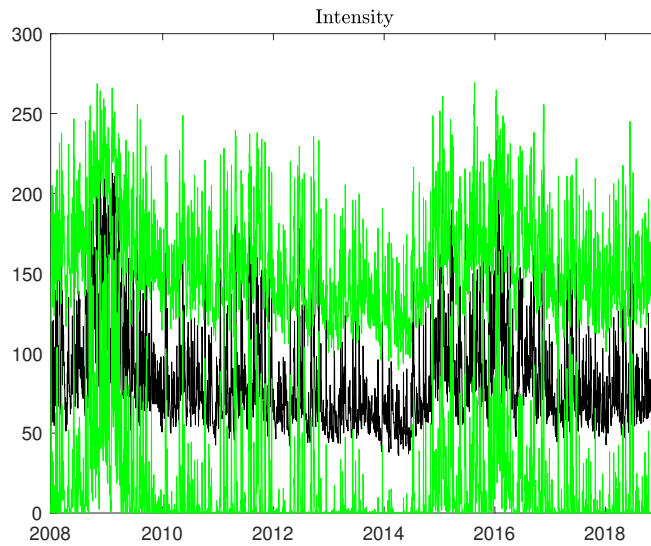


Figure 8: The figure presents the filtered jump intensity from model I. The posterior mean and (5, 95)% quantiles are reported at each time point.

Table 4 presents the parameter estimates for the two models obtained with the two-stage SMC sampler. As a first remark, when self-excitation is considered, volatility is much more persistent. Indeed, k is equal to 21.0876 (1.9348) in Model I and 2.8190 (0.5036) in Model II. One possible explanation for this is that much of the variation is due to the self-exciting jump intensity, dampening the contribution of volatility. However, the vol-of-vol parameter σ_v is equal to 0.426 (0.1134) in Model I and 0.8303 (0.0719) in Model II, confirming that volatility dynamics in oil market is clearly stochastic. Second, in line with the previous literature ([7, 32]), we find evidence of volatility jumps. In particular, the estimate for μ_v is 0.0213 (0.0023) in Model I and 0.1277 (0.0463) in Model II. Third, the posterior mean of the parameter controlling the self-exciting effect, α , is equal to 135.1930 (5.0244); then α is well identified and constitutes a key feature of the jump dynamics in the oil market. Fourth, the convenience yield dynamics is pretty much the same within the two models. The parameters are well identified and the convenience yield risk premium is positive and statistically significant, which confirms previous studies in the literature.

Furthermore, it is possible to compare the models by looking at the marginal likelihood and the log Bayes factor. In both cases we can say that Model I performs better than Model II. For instance, the log Bayes factor⁴ of Model I with respect to Model II is 9.1271, which means that Model I is decisively better than Model II in fitting the data.

⁴For any two given models, M_1 and M_2 , if the value of the log Bayes factor is between 0 and 1.1, M_1 is barely worth mentioning; if it is between 1.1 and 2.3, M_1 is substantially better than M_2 ; if it is between 2.3 and 3.4, M_1 is strongly better than M_2 ; if it is between 3.4 and 4.6, M_1 is very strongly better than M_2 ; and if it is larger than 4.6, M_1 is decisively better than M_2 .

	Model I		Model II	
Θ	Mean	Std	Mean	Std
μ	0.0699	0.0772	0.0109	0.0691
μ_J	-0.0017	0.0042	-0.0118	0.0827
σ_J	0.0287	0.0016	0.1104	0.0537
k	21.0876	1.9348	2.8190	0.5036
$\bar{V}$	0.0055	0.0028	0.1279	0.0134
σ_v	0.4261	0.1134	0.8303	0.0719
ρ_v	-0.6206	0.1607	-0.4161	0.0502
γ	0.6708	0.0041	0.6721	0.6721
$\bar{\delta}$	0.2304	0.0086	0.2639	0.0079
σ_δ	1.7690	0.0183	1.7646	0.0197
β	139.8989	4.7163	(—)	(—)
α	135.1930	5.0244	(—)	(—)
φ_δ	0.1272	0.0052	0.1500	0.0048
μ_v	0.0213	0.0023	0.1277	0.0463
MLLH	7.3678e+4		7.3022e+4	

Table 4: Parameter estimates for model I and II. For each parameter, the posterior mean and standard deviation are reported. The last row reports the marginal log likelihood estimated with the smooth particle filter.

Now, we can provide some additional insights regarding the convenience yield dynamics and the forecasting ability of the models in terms of futures pricing. In particular, with this kind of models it is common to observe a lack of fit in the filtered convenience yield process [8]. To check if our approach could give a reliable estimate of this variable, we follow [8] and compute the implied convenience yield using the estimated parameters together with the price of traded futures contracts $F(t, T_i)$. Then, we compare the filtered convenience yield from our smooth particle filter with the implied one. The result of this exercise is shown in Figure 9 and confirms that our filtering method is able to well reproduce the convenience yield dynamics implied from data (especially for longer maturities).

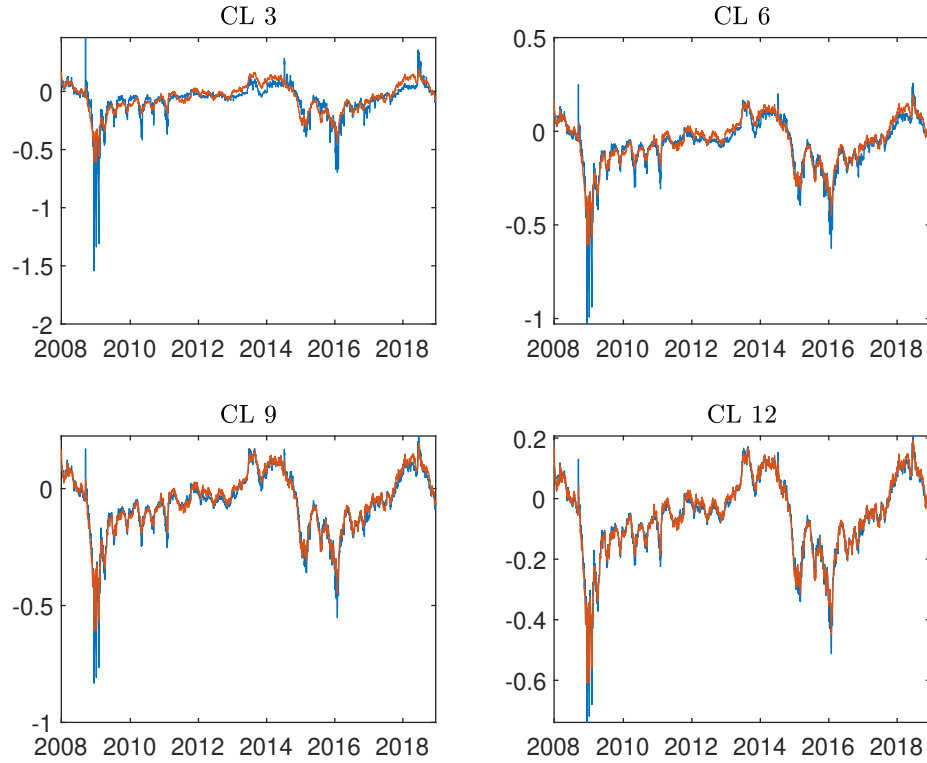


Figure 9: The figure compares the filtered convenience yield (orange) against the implied convenience yield (blue) computed using futures contracts with $\tau = \{3, 6, 9, 12\}$ and the posterior mean parameter estimates from Model I.

Second, we check the pricing errors of futures contracts when Model I and Model II are considered. For example, in Figure 10, we can see how the models behave in terms of futures pricing out of the sample, i.e. from 02/01/2019 to 17/01/2020 (263 observations). To this end we run our smooth PF on the new data using the parameters obtained during the previous estimation and then we filter out the convenience yield. As we can see the results are good with both models, with slightly better performances for Model I.

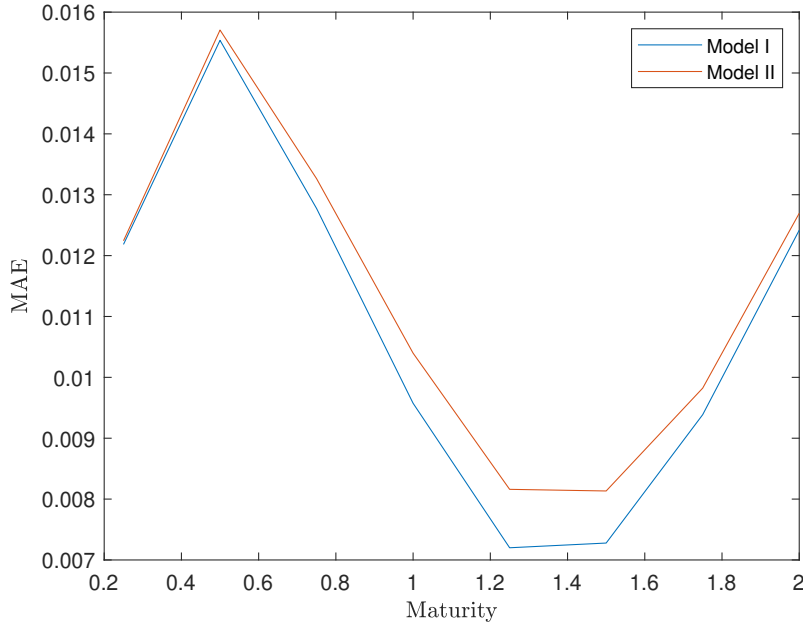


Figure 10: The figure presents the out of sample Mean Absolute Error (MAE) in terms of pricing futures contracts with Model I and Model II.

5 An Optimal Dynamic Hedging Strategy

Hedging in crude oil market is an important issue not only for producers, but also for energy traders and investors [6]. Indeed, price fluctuations lead to an increase in volatility, and so the risk coming from investing in the spot market need to be mitigated. In particular, the natural way to hedge a long (short) position in the spot market is to sell (buy) a certain number of futures contracts. The quantity of futures needed to cover a spot position is called *hedge ratio*. The determination of the optimal hedge ratio depends on the chosen objective function. In the literature, the hedge ratio is modeled as a time-varying variable (see [29, 30, 6, 10, 4, 2]), which minimizes the variance of the portfolio $\Pi = S - hF$, obtaining as solution

$$h_{t-1}^* = \frac{\text{Cov}_t(S, F)}{\text{Var}_t(F)}. \quad (32)$$

At this point, it is important to remark that our approach is different from that usually

considered in the literature. In particular, the most popular choice is given by discrete time models like multivariate GARCH models [10, 4] and Markov switching models [6, 2]. On the other hand, continuous time models have received much less attention and one example of such an approach is given by Liu et al. [30]. In their paper, related to industrial metals' market, they specify one dynamics for the spot price and one for the futures and then the estimation is carried out separately. However, our model provides a direct link between spot and futures, meaning that in our estimation framework we cannot avoid no arbitrage issues and we need to resort to a risk-neutral argument. In view of possible extensions of the present estimation method including different derivatives contracts, European options for example, the risk-neutral approach represents the most convenient and natural modelling framework. Also in the present setting, the risk-neutral approach provides the necessary consistency relations between spot and futures prices.

Now, let us properly specify the hedging portfolio. In particular, the log-spot price is given by:

$$dX_t = \left(r - \frac{1}{2}V_t - \mu^* \lambda_t - \delta_t \right) dt + \sqrt{V_t} dW_{x,t} + dJ_t^x.$$

Moreover the latent states are described by the following equations:

$$\begin{aligned} dV_t &= k(\bar{V} - V_t)dt + \sigma_v \sqrt{V_t} \left(\rho dW_{x,t} + \sqrt{1 - \rho^2} dW_{v,t} \right) + dJ_t^v, \\ d\delta_t &= \gamma(\bar{\delta} - \delta_t)dt + \sigma_\delta dW_{\delta,t}, \\ d\lambda_t &= \beta(\lambda_\infty - \lambda_t)dt + \alpha dN_t. \end{aligned}$$

Then, By applying Itô's lemma on the log-futures pricing function $f(X, V, \lambda, \delta, t, T) = X_t + A(\tau) + C(\tau)\delta_t$ we get the dynamics of Y_t :

$$\begin{aligned} dY_t &= \left[r - \frac{1}{2}V_t - \lambda_t \mu^* - \delta_t - A'(\tau) - C'(\tau)\delta_t + C(\tau)\gamma(\bar{\delta} - \delta_t) \right] dt \\ &\quad + \sqrt{V_t} dW_{x,t} + C(\tau)\sigma_\delta dW_{\delta,t} + dJ_t^x \end{aligned}$$

Then, the MV hedging ratio is given by

$$h^* = \frac{V_t + (\mu_J^2 + \sigma_J^2)\lambda_t}{V_t + C^2(\tau)\sigma_\delta^2 + (\mu_J^2 + \sigma_J^2)\lambda_t}. \quad (33)$$

Additional details are given in subsection 5.1.1 and in Appendix A, where we also introduce

a skewness-type objective function.

5.1 Empirical Application

In our application we follow previous studies (e.g. [6, 30]) and consider as hedging instrument the most liquid futures, i.e. the one with shortest maturity, which in our data set is the CL3 contract. Thus, we compute a daily hedge ratio according to the MV approach. In Figure 11 the optimal time varying hedge ratio h_t^* computed with Model I and Model II is shown. The time varying hedge ratios indicates that the portfolio should be frequently re-balanced as the market conditions changes. In particular, during the global financial crisis (2008-2010) we observe a positive jump in the hedge ratio, which means that investors are more careful and prefer to hedge more. Then, from 2011 to 2014 the market is less volatile and the hedge ratios are smaller providing to the hedgers a lower exposition on futures. Finally, from 2015-2017 we observe again an increase in the optimal hedging strategy due to another period of market turmoil. Moreover, we provide some descriptive statistics about the hedge ratios in Table 5.

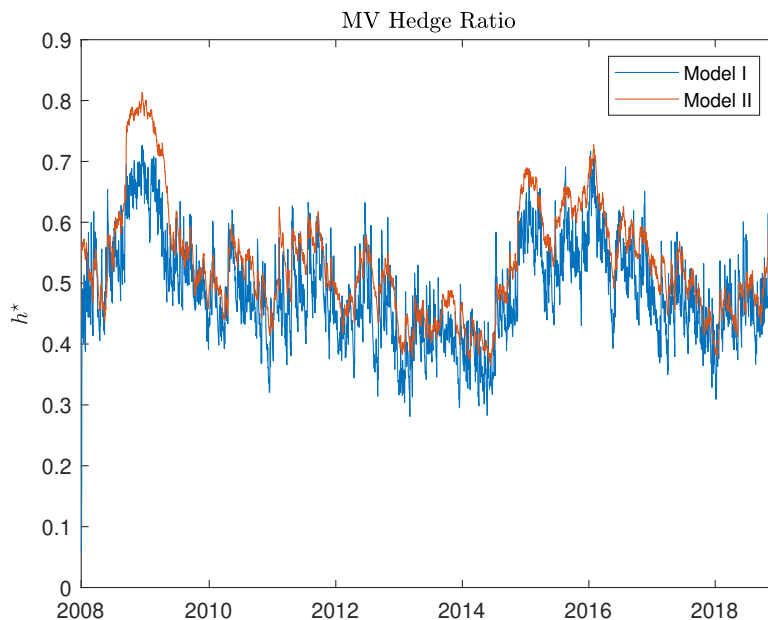


Figure 11: The figure presents the optimal MV hedge ratio time series for Model I (blue) and Model II (orange).

	Mean	Std Dev	Skewness	Kurtosis	Min	Max
MV Hedge I	0.4869	0.0833	0.2863	3.0156	0.0595	0.7263
MV Hedge II	0.5337	0.0904	0.8253	3.5483	0.3648	0.8136

Table 5: MV Hedge ratio statistics.

In the literature it is common to evaluate a particular hedging strategy in terms of variance reduction and utility improvements with respect to the un-hedged position [29, 2, 4]. Therefore, we consider the following measure of hedging effectiveness

$$HE_1 = 1 - \left[\frac{\text{Var}(\Pi_h)}{\text{Var}(\Pi_{un})} \right], \quad (34)$$

where, the hedging portfolios Π_h are computed using percentage log-returns of spot and futures and the corresponding optimal hedge ratios. Hence, we are evaluating the variance reduction with respect to the un-hedged portfolio Π_{un} , which is formed only by the spot position. If $HE_1 = 0$ we do not reduce risk at all, whereas $HE_1 = 1$ imply a 100% reduction in the variance. According to our previous findings about the hedge ratio variability we consider the whole sample and two different sub-periods: 2008-2010 and 2015-2017. In this way we are able to assess the hedging performances even during turbulent periods. The results are shown in Table 6.

Interval	Model I	Model II
2008-2018	0.6832	0.7270
2008-2010	0.6627	0.7100
2015-2017	0.7176	0.7610

Table 6: The table presents the variance reduction with respect to the un-hedged portfolio.

Overall, we obtain clear improvements with respect to the un-hedged position. For instance, during the whole sample, we reduce the portfolio variance by 68.32% with Model I and by 72.70% with model II. Moreover, even during periods of financial turmoil we are able to reduce significantly the variance of the portfolio. This confirms the importance of hedging for producers and investors.

According to [2], hedging effectiveness is more appropriately assessed by considering the economic benefits from hedging using the hedger's utility function. Then, if ξ represents the risk aversion of an investor and R_h is the return on the hedged portfolio, the utility function is given by

$$\mathbb{E}[U(R_h)] = \mathbb{E}[R_h] - \xi \text{Var}[R_h]. \quad (35)$$

By assuming that expected returns from the hedged portfolio are equal to zero and the degree of risk aversion is 4⁵ we compute the realized utility for Model I, Model II and the un-hedged portfolio in Table 7.

Interval	Model I	Model II	Un-hedged
2008-2018	-7.5929	-6.5421	-23.9646
2008-2010	-18.0564	-15.5256	-53.5398
2015-2017	-7.7348	-6.5467	-27.3873

Table 7: The table presents the realized MV utility for Model I, Model II and the un-hedged portfolio.

Now, some comments are needed. First, adopting the optimal hedging strategy we obtain substantial utility improvements with respect to the un-hedged position, no matter which sub-period is considered. Second, during the global financial crisis we get the worst results due to increased portfolio variance. Third, despite the better performance in term of Bayes factor, Model I is performing slightly worse than Model II, both regarding variance reduction and utility improvement.

In the next subsection we are going to explore if the inclusion of high order effects could give some improvements in the present hedging application.

5.1.1 Higher Order Hedging

The presence of jumps and stochastic volatility in the oil market raises a question about the adequacy of a minimum variance (MV) objective function. Indeed, by adopting the MV approach we are not fully exploiting the distributional properties of these models, i.e. we are neglecting higher order effects, which could influence the hedging ratio. Furthermore, the relevance of skewness in characterizing risk preferences has been pointed out in a rich amount of literature: we mention the papers by Post et al. [40], by Chiu [11], by Dahlquist et al. [13] and by Kraus and

⁵This assumptions are in line with most empirical studies as [29, 2].

Litzenberger [28]. Therefore, we consider the minimization of the following objective function:

$$\min_h [\text{Var}_t(\Pi_t) - \eta \text{Asy}_t(\Pi_t)], \quad (36)$$

where, $\Pi_t = X_t - hY_t$ is the portfolio formed by $X_t = \log(S_t)$ and $Y_t = \log(F_{t,T})$; η is a constant risk-aversion parameter and $\text{Asy}[x] = \mathbb{E}[(x - \mathbb{E}[x])^3]$.

In order to compute the optimal hedging ratio we need $\text{Var}_t[\Pi_t]$ and $\text{Asy}_t[\Pi_t]$. Hence, in analogy with [30] we can proceed by computing the instantaneous conditional moments. The detailed derivation of the hedging ratio is given in Appendix A. Once we have obtained the expression for the hedging portfolio we take the first derivative with respect to h and set it equal to zero. In the end we get two solutions for h :

$$\begin{aligned} h^* = \pm \frac{1}{2(3\eta\lambda_t\mu_J^3 + 9\eta\lambda_t\mu_J\sigma_J^2)} & \left((2C^2\sigma_\delta^2 - 6\eta\lambda_t\mu_J^3 - 18\eta\lambda_t\mu_J\sigma_J^2 + 2\lambda_t\mu_J^2 + 2\lambda_t\sigma_J^2 + 2V)^2 \right. \\ & - 4(3\eta\lambda_t\mu_J^3 + 9\eta\lambda_t\mu_J\sigma_J^2)(3\eta\lambda_t\mu_J^3 + 9\eta\lambda_t\mu_J\sigma_J^2 - 2\lambda_t\mu_J^2 - 2\lambda_t\sigma_J^2 - 2V_t) - 2C^2\sigma_\delta^2 + 6\eta\lambda_t\mu_J^3 \\ & \left. + 18\eta\lambda_t\mu_J\sigma_J^2 - 2\lambda_t\mu_J^2 - 2\lambda_t\sigma_J^2 - 2V_t \right)^{1/2}. \end{aligned} \quad (37)$$

Since crude oil spot and futures are positively correlated, from now on, we shall consider only the positive solution of (37). As in the previous section we compute daily hedge ratios according to our optimal strategy. Nevertheless, in Figure 12 we observe a surprising result: the MV and the Asy-Variance hedging strategies give substantially the same hedge ratio. This means that the contribution given by the third-order moment is negligible in computing the optimal hedging strategy (especially for Model I).

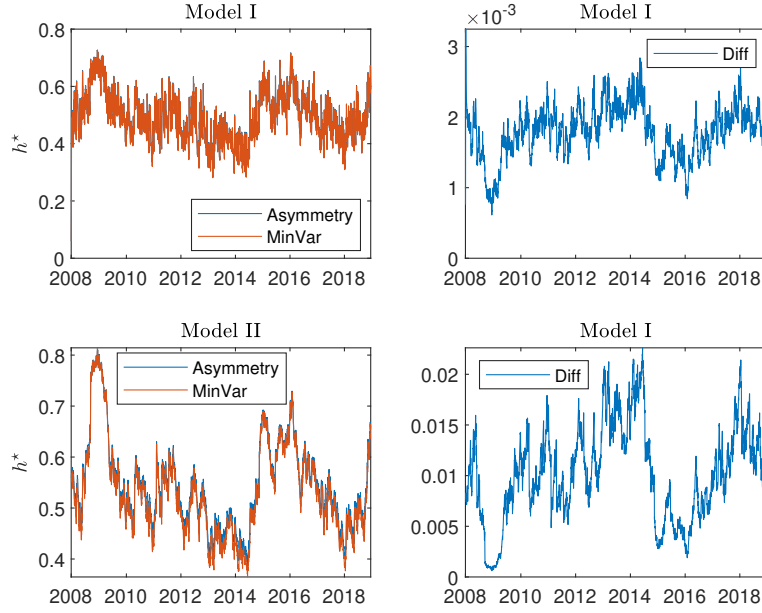


Figure 12: The figure presents in the left panels the optimal MV and Asy-Variance hedge ratios time series for Model I (blue) and Model II (orange). In the right panels we plot the difference between the two strategies.

Given this result, we include the skewness instead of the third order central moment; then the optimization problem reads as follows

$$\min_h [\text{Var}_t(\Pi_t) - \eta \text{Skew}_t(\Pi_t)], \quad (38)$$

where,

$$\text{Skew}(x) = \frac{\mathbb{E}[(x - \mathbb{E}[x])^3]}{\text{Var}(x)^{3/2}}.$$

In this way we are not able to get a closed form solution, then a numerical minimization is adopted. Hence, the hedge ratio obtained numerically is compared to the MV hedging ratio in Figure 13, from which it is clear that the contribution of skewness is not negligible. In the sequel we denote the Variance-Skewness hedging as VS.

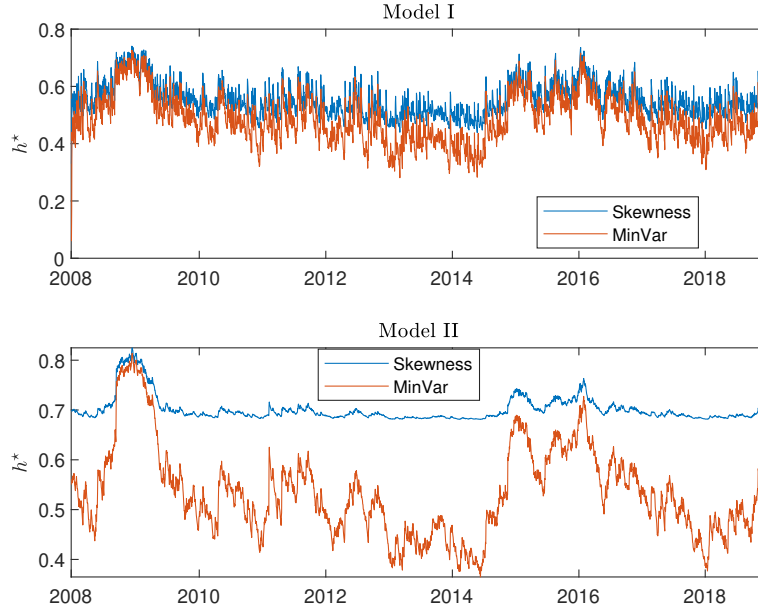


Figure 13: The figure presents the optimal MV and VS hedge ratios time series for Model I (blue) and Model II (orange).

Now, in order to investigate the performance of this new strategy we need to consider a metric which is coherent within the optimizations goals. Then, we resort again to the realized utility:

$$\mathbb{E}[U(R_h)] = -\text{Var}[R_h] + \eta \text{Skew}[R_h], \quad (39)$$

where, η is the risk aversion parameter related to skewness and it is set equal to 4. The result of this computation is given in Table 8.

Interval	Model I	Model II	Un-hedged
2008-2018	0.9478	3.5841	-5.5189
2008-2010	-1.0194	1.4152	-12.6846
2015-2017	-0.4642	0.6407	-5.7923

Table 8: The table presents the realized VS utility for Model I, Model II and the un-hedged portfolio.

From the results we infer that the realized utility is giving very good performances. In

particular, Model II gives a positive realized utility in every scenario. The explanation stems from the fact that positive skewness is increasing the utility of the investor.

To further investigate this finding we test the hedging strategies out of sample, i.e. we consider 263 observations from 02/01/2019 to 17/01/2020. Hence, we filter out the state variable by running our smooth particle filter using the parameters estimated in the sample. Then, we compute the realized utility of each strategy. The results are shown in Table 9. In this way we can appreciate the performance of each strategy with respect to the un-hedged position, by matching their respective optimization goals.

Interval	Model I	Model II	Un-hedhged
MV	-3.8155	-3.1600	-17.9647
VS	0.1262	2.5119	-2.5263

Table 9: The table presents the out of sample MV and VS realized utility for Model I and Model II. The realized utility of the un-hedged portfolio is also reported.

The results indicate that both strategies give better results with respect to the un-hedged portfolio. In particular, the VS strategy provides positive utility, no matter which model is considered. Again, we observe that Model II performs better than Model I.

By resuming the previous results, it seems that the inclusion of high order effects lead to substantial improvements in the present hedging application with respect to the usual MV approach. However, the self-exciting feature of Model I does not provide better performances with respect to Model II. Model I actually includes one more essential source of risk to hedge with respect to Model II, constituted by the random nature of the jump intensity, which produces unpredictable clusters of jumps, making hedging more difficult to realize. A systematic and comprehensive description of other possible hedging strategies in the context of self-exciting jump-diffusion models for oil price dynamics is outside the purpose of this paper and will be the subject of future investigations.

6 Concluding Remarks

In this paper we propose a jump-diffusion model for oil price dynamics including self-exciting effects. Our model includes a stochastic dynamics for both the volatility and the yield coefficient. We provide an estimation method of Bayesian type based on a suitable adaptation of a SMC

method proposed by Fulop and Li [22]. Our results show that the introduction of a jump intensity of Hawkes type improves the forecasting ability of this model with respect to a model without self-exciting effects. Moreover its affine feature allows to compute explicitly futures contracts prices providing a setting for accurate parameters estimation. Finally we compute an optimal hedging strategy based on futures trading. This optimal hedging strategy is obtained not only by minimizing variance, but by taking into account higher order moments, i.e. skewness. This optimal hedging strategy exhibits some interesting features and gives to the hedger better results with respect to the most popular MV approach.

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A Optimal Hedging Solution

In this appendix we provide detailed calculations of the optimal hedge ratio. Let us start from the portfolio variance:

$$\frac{1}{dt} \text{Var}_t[d\Pi_t] = \frac{1}{dt} \left(\text{Var}_t[dX_t] + h^2 \text{Var}_t[dY_t] - 2h \text{Cov}_t[dX_t, dY_t] \right).$$

consider the first term:

$$\begin{aligned} \frac{1}{dt} \text{Var}_t[dX_t] &= \frac{1}{dt} \text{Var}_t \left[\left(r - \frac{1}{2} V_t - \mu^* \lambda_t - \delta_t \right) dt + \sqrt{V_t} dW_{x,t} + dJ_t^x \right], \\ &= V_t + \frac{1}{dt} \text{Var}_t [J_x dN_t], \end{aligned}$$

where,

$$\begin{aligned} \text{Var}_t [J_x dN_t] &= \mathbb{E}[J_x^2 dN_t^2] - (\mathbb{E}[J_x dN_t])^2 = \mathbb{E}[J_x^2] \mathbb{E}[dN_t] - (\mathbb{E}[J_x] \mathbb{E}[dN_t])^2, \\ &= (\mu_J^2 + \sigma_J^2) \lambda_t dt - \mu_J \lambda_t^2 dt^2 = (\mu_J^2 + \sigma_J^2) \lambda_t dt. \end{aligned}$$

Therefore we have

$$\frac{1}{dt} \text{Var}_t[dX_t] = V_t + (\mu_J^2 + \sigma_J^2) \lambda_t.$$

The next object is the instantaneous variance of log-futures:

$$\begin{aligned} \frac{1}{dt} \text{Var}_t[dY_t] &= \frac{1}{dt} \text{Var}_t \left[\sqrt{V_t} dW_{x,t} + C(\tau) \sigma_\delta dW_{\delta,t} + J_x dN_t \right], \\ &= V_t + C^2(\tau) \sigma_\delta^2 + (\mu_J^2 + \sigma_J^2) \lambda_t. \end{aligned}$$

The covariance between X and Y is given by

$$\begin{aligned} \frac{1}{dt} \text{Cov}_t[dX_t, dY_t] &= \frac{1}{dt} \text{Cov}_t \left[\sqrt{V_t} dW_{x,t} + J_x dN_t, \sqrt{V_t} dW_{x,t} + C(\tau) \sigma_\delta dW_{\delta,t} + J_x dN_t \right], \\ &= V_t + (\mu_J^2 + \sigma_J^2) \lambda_t. \end{aligned}$$

Then we obtained the conditional variance of the portfolio:

$$\frac{1}{dt} \text{Var}_t[d\Pi_t] = V_t + (\mu_J^2 + \sigma_J^2) \lambda_t + h^2 \left[V_t + C^2(\tau) \sigma_\delta^2 + (\mu_J^2 + \sigma_J^2) \lambda_t \right] - 2h \left[V_t + (\mu_J^2 + \sigma_J^2) \lambda_t \right].$$

Now we consider the (non-standardized) third moment:

$$\frac{1}{dt}\mathbb{E}_t \left[(d\Pi - \mathbb{E}(d\Pi))^3 \right] = \frac{1}{dt}\mathbb{E}_t \left[d\Pi^3 - (\mathbb{E}(d\Pi))^3 - 3d\Pi^2\mathbb{E}(d\Pi) + 3d\Pi(\mathbb{E}(d\Pi))^2 \right].$$

The first element can be computed as follows

$$\begin{aligned} \frac{1}{dt}\mathbb{E}_t [d\Pi^3] &= \frac{1}{dt}\mathbb{E}_t [dX^3 - h^3 dY^3 - 3hdX^2 dY + 3h^2 dX dY^2], \\ &= (\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t - h^3(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t - 3h(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t + 3h^2(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t. \end{aligned}$$

Now define,

$$\begin{aligned} \mu_X &= r - \frac{1}{2}V_t - \mu^*\lambda_t - \delta_t, \\ \mu_Y &= r - \frac{1}{2}V_t - \lambda_t\mu^* - \delta_t - A'(\tau) - C'(\tau)\delta_t + C(\tau)\gamma(\bar{\delta} - \delta_t). \end{aligned}$$

Then we compute,

$$\begin{aligned} \mathbb{E}_t [d\Pi] &= \mathbb{E}_t [dX] - h\mathbb{E}_t [dY], \\ &= \mu_X dt + \mu_J\lambda_t dt - h(\mu_Y dt + \mu_J\lambda_t dt). \end{aligned}$$

given this we have that $(\mathbb{E}_t [d\Pi])^3 = (\mathbb{E}_t [d\Pi])^2 = \mathbb{E}_t [d\Pi^2\mathbb{E}_t [d\Pi]] = 0$. Therefore,

$$\begin{aligned} \frac{1}{dt}\mathbb{E}_t \left[(d\Pi - \mathbb{E}(d\Pi))^3 \right] &= \frac{1}{dt}\mathbb{E}_t [d\Pi^3], \\ &= (\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t - h^3(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t - 3h(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t + 3h^2(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t. \end{aligned}$$

Now, we take the derivative of the objective function with respect tot h,

$$\begin{aligned} \text{Var}_t(\Pi_t) - \eta\text{Asy}_t(\Pi_t) &= V_t + (\mu_J^2 + \sigma_J^2)\lambda_t + h^2 [V_t + C^2(\tau)\sigma_\delta^2 + (\mu_J^2 + \sigma_J^2)\lambda_t] - 2h [V_t + (\mu_J^2 + \sigma_J^2)\lambda_t] \\ &\quad - \eta [(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t - h^3(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t - 3h(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t + 3h^2(\mu_J^3 + 3\mu_J\sigma_J^2)\lambda_t]. \end{aligned}$$

If we set this quantity equal to zero and solve for h we get (37).